SURVEY & RESEARCH METHODS

Kirschner Awonatey Amaanya

Introduction

In our quest to achieve greater heights in all our endeavours, especially in this technological age, we are often confronted with problems. Abstract arguments cannot bring out solution to these problems or pave way for discoveries and innovations or establish a point effectively. Serious efforts must therefore be made to make new discoveries and innovations based on an existing theories, discoveries and inventions. These can be established by making actual observations, experiments and serious "studies" to a body of consistent point called theories, laws or concepts. A way to achieve these is by research.

RESEARCH DESIGNS

Research is therefore the ultimate source of new knowledge, laws and applications. Research can be defined as a planned enquiry to investigate an issue that might have risen (or any organized inquiry carried out to provide information for the solution of a problem. Research contributes tremendously to economic development in diverse ways. It serves as a basis of scientific development and a tool for policy formulation.

Research leads to discovery, for example in medicine, where research work and their findings are geared towards seeking vaccine to cure HIV/AIDS menace. Some research findings might give results/information which may be seen ignorant. Research work in the form of a project report, thesis/dissertation and/or publication may be a requirement for an award of a degree or diploma to students, and also for personnel to gain appointment or promotion in some research institutions.

Census, Sample Survey and Survey Errors

There are two types of survey research, namely census and sample survey. A census is a complete count or enumeration of the entire population. A sample survey is the study of a portion (or partial enumeration) of a population for the purpose of making estimated assertion about the population. A census is therefore a 100% survey.

At various stages of a survey being conducted errors occur. Some of these errors may be random which can occur to produce smaller or larger estimates at any pattern. These estimates produce a net result where they cancel themselves out. These errors are sometimes called accidental, occurring as a result of a multiplicity of causes rather than due to particular assignable sources.

An error in a survey is the deviation from the correct or true value. There are two types of errors that affect results of surveys. These are sampling and non-sampling errors.

Sampling Errors

The sampling error is a measure of the variability between all possible samples. These errors arise solely from the random sampling variation that are present when n(sample size) of sample units are measured instead of the complete population of N units.

The standard error of a particular statistic (like sample mean or proportion) is the measure of the magnitude of the sampling error which is the square root of the variance of the sampling distribution of the statistic. The standard error can be used to calculate confidence intervals which apart from non-sampling error, the true value for the population can reasonably be assumed to fall.

The sampling errors are reduced if the sample size is sufficiently large.

Non-Sampling Errors (NSE): Overview

Non-sampling errors (NSE) are any phenomena other than sampling errors that cause a difference between a population parameter and its estimate. These errors or biases can occur at all stages of the planning and conduct of a survey.

Non-sampling errors pose serious danger to survey results because their magnitude is often large and there is no satisfactory statistical theory or technique for measuring them accurately. Unlike sampling errors, they do not necessarily decrease with an increase in sample size.

Sources of Non-Sampling Errors

Major causes of non-sampling errors include:

- Errors in sampling operations
- Non-response problems
- Lack of knowledge and understanding of survey concepts
- Concealment of truth by respondents
- Operationalization or conceptual problems
- Improper design of questionnaire
- Measurement errors
- Wrong processing of data and interpretation of results
- Inadequate supervision of fieldwork

Effects of Non-Response Errors

Non-response occurs when selected units fail or refuse to provide information. Its major effects on survey results include:

- Reduction in the number of units covered and amount of data available
- Increase in the standard error of estimates
- Biased results when respondents and non-respondents differ systematically
- Predictions and conclusions that may be misleading

Non-response bias may persist regardless of effort made to increase response rate.

Control of Non-Sampling Errors

Although non-sampling errors cannot be completely avoided, they can be controlled through:

- Proper and careful survey design
- Reduction of non-response through follow-up and substitution
- Good questionnaire design and data collection procedures
- Adequate supervision and inspection of fieldwork
- Pilot surveys before the main survey
- Post-enumeration surveys (PES) to detect coverage and content errors

Sampling Issues: Overview

There are several ways researchers can select samples. One of the most useful, but also the most difficult, methods is random sampling. In statistics, the term random does not mean haphazard; rather, it means that every member of the population has a known chance of being selected into the sample.

The main goal of proper sampling is to obtain a sample that allows valid generalization of results to the population of interest.

In random sampling, every member of the population has a chance of being selected. The major advantage of random sampling is that differences between the sample and the population are not systematic.

Although randomly selected samples may differ from the population, especially when the sample size is small, these differences occur by chance rather than by bias in the selection process.

Representative sampling involves purposely selecting cases so that the sample matches the population on important characteristics such as sex, age, occupation or region.

For example, if 55% of adults in the public sector are male, then 55% of the sample should be male. This method increases the likelihood that results can be generalized to the population, although it may be costly and time-consuming

Convenience sampling involves selecting participants based on ease of access, proximity, and willingness to participate.

Although convenience sampling is less labor-intensive than random or representative sampling, it is acceptable if the selected sample does not differ from the population in ways that influence the study outcomes.

Data are factual information obtained from observations, measurements, or records collected from a sample. Data are not limited to numbers and may include qualitative information such as sex, age, or categories.

Data involving more than one variable are called multivariate data, while data involving two variables are referred to as bivariate data.

Sources of Data

Data used in statistical studies are classified as **primary** or **secondary** depending on whether they are collected specifically for the study.

Primary Data are collected under the control and supervision of the researcher for a specific purpose, usually through surveys or experiments. Their main advantages are accuracy, relevance, and control over data collection. However, primary data are often expensive and time-consuming to collect.

Secondary Data are collected by other individuals or organizations for purposes different from the current study. They may be obtained from reports, published statistics, surveys, or administrative records. Secondary data are cheaper, faster to obtain, and useful for general reference and preliminary investigations, but they may be outdated, unreliable, or unsuitable for the researcher's specific needs.

Data Collection Methods: Overview

Data collection is a fundamental step in any statistical investigation. Numerical data provide the raw material needed for analysis, planning, forecasting, decision making, policy formulation, administration, and scientific theory development.

The basic methods of data collection in research include:

- Abstraction from published or unpublished records
- Conducting surveys
- Direct observation

Survey Methods of Data Collection

A survey involves collecting data using a structured questionnaire designed to obtain information relevant to a study. Survey data are collected mainly through:

- Personal interviews
- Telephone interviews
- Self-administered questionnaires

Direct Observation and Questionnaire Design

Direct observation involves systematically watching and recording actual behaviour or events as they occur. It provides accurate, real-time data but may be expensive, time-consuming, and subject to observer bias.

Good questionnaire design is essential for quality data collection and should emphasize:

- Clear objectives and instructions
- Simplicity and brevity
- Logical sequence of questions
- Avoidance of sensitive or misleading questions
- Proper pre-testing through pilot surveys

Unit 2: Sampling Methods and Concepts

Sampling is the process of learning about a population by studying a subset (sample) drawn from it. Data obtained from a sample are used to estimate population quantities such as means, proportions, or totals, known as parameters.

Although population parameters can be determined exactly through a census, sampling is preferred because it is less costly, faster, and often more accurate than complete enumeration.

Sampling is used because:

- It provides reliable information at lower cost
- Results can be obtained more quickly
- Sample surveys may be more accurate than censuses

Key sampling concepts include:

- Sampling units (elementary and primary)
- Sampling frame
- Target population

Non-Probability Sampling

Non-probability sampling methods do not involve random selection, and the probability of selecting an element cannot be determined. As a result, sampling errors cannot be measured and reliable statistical inference is not possible.

Common non-probability sampling methods include:

- Convenience sampling
- Judgmental (purposive) sampling
- Quota sampling

These methods are easy and cost-effective but are subject to bias.

Probability (Random) Sampling

Probability sampling selects samples such that every element in the population has a known and non-zero chance of selection. This ensures that the sample is representative and free from systematic bias.

Probability sampling forms the basis of statistical inference. Common methods include:

- Simple random sampling
- Stratified sampling
- Systematic sampling
- Cluster and multistage sampling

Simple Random Sampling

Simple random sampling selects sample units such that each element in the population has an equal chance of selection. Selection is done without replacement and without regard to order.

Advantages:

- Eliminates personal bias
- Allows verification of efficiency of estimates

Disadvantages:

- Requires a complete and up-to-date sampling frame
- Can be costly and time-consuming
- May produce unbalanced samples for small populations
- Often requires larger sample sizes than stratified sampling

Stratified Sampling

Explanation:

Stratified sampling divides the population into homogeneous, non-overlapping subgroups (strata) based on a stratifying factor such as sex, age, income, or education. A simple random sample is then drawn from each stratum, and all sub-samples are combined to form the overall sample.

Pros:

- More representative than simple random sampling
- Higher accuracy and precision of estimates
- Reduced sampling error
- Useful for large, diverse populations
- Administratively convenient

Cons:

- Requires detailed population information
- More complex to design and implement
- Incorrect stratification reduces efficiency

Systematic Sampling

Explanation:
Systematic sampling selects every k-th unit from a population list after choosing a random starting point (pivot). It is often used when a population is ordered or listed.

Pros:

- Simple and easy to implement
- Less time-consuming and less costly
- More efficient than simple random sampling if list is random
- Ensures good spread across the population

Cons:

- Can be biased if population has periodic patterns
- Units are not independent
- Sampling error cannot be easily estimated
- Limited number of possible samples

Cluster Sampling

Explanation:
Cluster sampling divides the population into clusters (often geographic). A random sample of clusters is selected, and either all units within selected clusters or a sample of units within them is studied.

Pros:

- Cost-effective, especially for geographically dispersed populations
- Operationally convenient
- Requires less detailed population lists

Cons:

- Less precise than stratified sampling
- Inefficient if clusters are heterogeneous
- Unequal cluster sizes reduce accuracy

Multi-Stage Sampling

Explanation:
Multi-stage sampling is an extension of cluster sampling carried out in several stages (e.g., region $\rightarrow$ district $\rightarrow$ community $\rightarrow$ household). Random sampling is applied at each stage.

Pros:

- Greatly reduces cost and fieldwork
- Suitable for large-scale national surveys
- Administrative convenience
- Allows coverage of large geographic areas

Cons:

- More complex to design and analyze
- Less efficient than single-stage sampling
- Sampling error accumulates across stages

Summary Comparison

Explanation:
Different sampling methods are chosen based on population structure, cost, time, and required accuracy.

Key Idea:

- Stratified sampling $\rightarrow$ best accuracy
- Systematic sampling $\rightarrow$ simplicity and speed
- Cluster sampling $\rightarrow$ cost efficiency
- Multi-stage sampling $\rightarrow$ large-scale surveys

Trade-off:
Higher accuracy usually comes with higher complexity and cost.

Notations and Definitions in Sample Surveys

Notations and Definitions In sample surveys, we measure and record for every unit that comes into the sample. These measurements are denoted by lowercase letters. Uppercase letters are used for population measurements.

Estimation of Sample Size

In the planning of a sample survey, a stage is always reached at which a decision must be made about the size of the sample (n). This decision is perhaps the most crucial stage. A sample that is too large implies a waste of resources, while a sample that is too small reduces the accuracy of the results. The decision cannot always be made satisfactorily, as often we do not possess enough information to be sure that our choice of n is optimal. The theory of sampling provides a framework within which to think intelligently about this problem.

The following example illustrates the steps involved in determining a suitable sample size:

Example: A researcher plans to study the inhabitants of an island. Among other things, he wishes to estimate the proportion of inhabitants belonging to blood group O. Co-operation has been secured so that it is feasible to take a simple random sample. The question is: How large should the sample be?

This question cannot be answered without first answering another: How accurately does the researcher wish to know the proportion of people with blood group O? He may state that he will be content if the proportion is correct within $\pm 5\%$, in the sense that if the sample gives a proportion of 0.43, then the true proportion of blood group O is sure to lie between 0.38 and 0.48.

To avoid misunderstanding, it may be advisable to point out to the researcher that accuracy within 5% cannot be absolutely guaranteed except by measuring everyone in the population. However, if a large n is taken, there is still a chance of obtaining a very unlucky sample that is in error by more than the desired 5%. The researcher replies that he is aware of this and is willing to take a 1 in 20 chance of getting an unlucky sample, and that all he asks for is the value of n, rather than a lecture on statistics.

Principal Steps in Choosing a Sample Size

1. There must be a statement concerning what is expected of the sample. This statement may be in terms of desired limits of error or precision. It is the responsibility of the personnel who wish to use the results of the survey, though they often need guidance in expressing their wishes numerically.
2. An equation connecting n with the desired precision of the sample must be found. The form of the equation varies with the statement of precision and the type of sampling contemplated. Probability sampling enables this equation to be constructed.
3. The equation contains certain parameters (unknown properties of the population) that must be estimated to give specific results.
4. Often, data are to be published for major subdivisions of the population, and desired limits of error are set for each subdivision. A separate calculation is made for n in each subdivision, and the total n is found by addition.

1. Usually more than one characteristic is measured in a survey (sometimes many items). If a desired degree of precision is prescribed for each item, the calculations lead to a series of conflicting values of n, one for each item. A method must be found for reconciling these values.
2. The value of n obtained must be appraised to see whether it is consistent with the resources available. This requires estimating the cost, labor, time, and materials required. It may become apparent that n must be drastically reduced. A hard decision must then be faced: whether to proceed with a much smaller sample size, reducing precision, or to postpone the survey until more resources are available.

Computation of Sample Size for Estimating a Proportion

Let d be the margin of error in estimating P (the proportion of units in class C), and α the small risk we are prepared to take that the actual error exceeds d. Then:

$$\Pr(|p - P| \geq d) = \alpha$$

Equivalently, using the normal approximation:

$$\Pr(p - Z_\alpha \cdot se(p) \leq P \leq p + Z_\alpha \cdot se(p)) = 1 - \alpha$$

where the standard error of p is:

$$se(p) = \sqrt{\frac{PQ}{n} \cdot \frac{N - n}{N - 1}}$$

Here, p is assumed to be normally distributed, $Q = 1 - P$, N is the population size, and n is the sample size.
The desired degree of precision or margin of error d is:

$$d = Z_\alpha \cdot se(p) = Z_\alpha \sqrt{\frac{PQ}{n} \cdot \frac{N-n}{N-1}}$$

Solving for Sample Size n: Squaring both sides:

$$d^2 = Z_\alpha^2 \cdot \frac{PQ}{n} \cdot \frac{N-n}{N-1}$$

Rearranging:

$$n(N-1)d^2 = Z_\alpha^2 PQ(N-n)$$

$$n(N-1)d^2 + nZ_\alpha^2 PQ = NZ_\alpha^2 PQ$$

$$n\left[Z_\alpha^2 PQ + (N-1)d^2\right] = NZ_\alpha^2 PQ$$

$$\boxed{n = \frac{NZ_\alpha^2 PQ}{(N-1)d^2 + Z_\alpha^2 PQ}}$$

This is the exact formula including the finite population correction (fpc).

Large Population Approximation

If N is very large so that the fpc factor can be ignored:

$$n_0 \approx \frac{Z_\alpha^2 PQ}{d^2}$$

Here, n_0 is the initial estimate of the sample size.
If the fpc factor cannot be ignored, then:

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}} \approx \frac{n_0}{1 + \frac{n_0}{N}}$$

In practice, we first calculate n_0. If $n_0/N < 0.05$ (negligible), n_0 is a satisfactory approximation to n.

Sample Size for Continuous Data: Relative Error

Sometimes we wish to control the **relative error** r in estimating the population mean $\hat{y}$ or total Y, such that:

$$\Pr\left(\frac{|\hat{y} - Y|}{Y} \geq r\right) = \alpha$$

Equivalently:

$$\Pr(Y - rY \leq \hat{y} \leq Y + rY) = 1 - \alpha$$

Assuming y is normally distributed, the standard error is:

$$se(y) = \sqrt{\frac{s^2}{n} \cdot \frac{N-n}{N}}$$

Then, the desired degree of precision or relative error satisfies:

$$rY = Z_\alpha \cdot se(y) = Z_\alpha \sqrt{\frac{s^2}{n} \cdot \frac{N-n}{N}}$$

Sample Size for Continuous Data: Relative Error

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Assuming y is normally distributed, the standard error is:

$$se(y) = \sqrt{\frac{s^2}{n} \cdot \frac{N - n}{N}}$$

Desired Precision and Formula Setup

The desired degree of precision or relative error satisfies:

$$rY = Z_\alpha \cdot se(y) = Z_\alpha \sqrt{\frac{s^2}{n} \cdot \frac{N-n}{N}}$$

Squaring both sides:

$$(rY)^2 = Z_\alpha^2 \frac{s^2}{n} \frac{N-n}{N}$$

Rearranging:

$$n(N)r^2Y^2 = Z_\alpha^2 s^2 (N-n)$$

Sample Size Formula with FPC

Solving for n including finite population correction (fpc):

$$n\left(Z_\alpha^2 s^2 + Nr^2 Y^2\right) = NZ_\alpha^2 s^2$$

$$\boxed{n = \frac{NZ_\alpha^2 s^2}{Z_\alpha^2 s^2 + Nr^2 Y^2}}$$

If N is very large (fpc ignored):

$$n_0 = \frac{Z_\alpha^2 s^2}{(rY)^2} = \frac{Z_\alpha^2 s^2}{c^2 Y^2}$$

where $c = r$ is the coefficient of variation (CV) of the estimate.
Adjustment if n_0/N is not negligible:

$$n = \frac{n_0}{1 + \frac{n_0}{N}}$$

Sample Size for Absolute Error and Total

If we wish to control the **absolute error** d in estimating the mean:

$$\Pr(|\hat{y} - Y| \geq d) = \alpha$$

$$d = Z_\alpha \cdot se(y) = Z_\alpha \sqrt{\frac{s^2}{n} \cdot \frac{N-n}{N}}$$

For large N (fpc ignored):

$$n_0 = \frac{Z_\alpha^2 s^2}{d^2}$$

If fpc cannot be ignored, adjust:

$$n = \frac{n_0}{1 + \frac{n_0}{N}}$$

For estimating total population:

$$n_0 = \frac{Z_\alpha^2 N^2 s^2}{d^2}, \quad n = \frac{n_0}{1 + \frac{n_0}{N}}$$

If we are interested in estimating the **total population** Y, the sample size can be computed based on the desired absolute error d as follows:

$$n_0 = \frac{Z_\alpha^2 N^2 s^2}{d^2}$$

If the finite population correction (fpc) cannot be ignored, the adjusted sample size is:

$$n = \frac{n_0}{1 + \frac{n_0}{N}}$$

This formula ensures that the sample size accounts for both variability in the population and the desired precision in estimating the total.

Example 2.3: Estimating Blood Group Proportion

Question 1: Suppose we wish to estimate the percentage of blood groups of inhabitants on an island. Given:

$$\alpha = 0.05, \quad p = 0.40, \quad d = 0.05, \quad N = 3200, \quad Z_\alpha = 1.96$$

Compute the initial sample size:

$$n_0 = \frac{Z^2 PQ}{d^2} = \frac{1.96^2(0.4)(0.6)}{0.05^2} \approx 369$$

Adjusting for finite population:

$$n = \frac{n_0}{1 + \frac{n_0}{N}} = \frac{369}{1 + \frac{369}{3200}} \approx 331$$

Solution: Hence, the required sample size is $n \approx 331$.

Example 2.3: Estimating Owned Houses and Two-Car Households

Question 2: Population: $N = 4000$ houses. We wish to estimate:

- Percentage of owned houses with standard error ≤ 0.02
- Percentage of two-car owned households with standard error ≤ 0.01

Solution: For optimum sample size, take $p_1 = 0.5$ (owned houses) and $p_2 = 0.1$ (two-car households).

$$n_1 = \frac{Z^2 P_1 Q_1}{V_1^2} = \frac{(0.5)(0.5)}{0.02^2} = 625$$

$$n_2 = \frac{Z^2 P_2 Q_2}{V_2^2} = \frac{(0.1)(0.9)}{0.01^2} = 900$$

Finite population adjustment:

$$n = \frac{n_0}{1 + \frac{n_0}{N}} = \frac{900}{1 + \frac{900}{4000}} \approx 735$$

Hence, choose $n \approx 735$ to satisfy both aims.

Example 2.3: Estimating Total Signatures

Question 3: In a population of petition sheets, estimate the total number of signatures with a margin of error $d = 1000$ and $\alpha = 0.05$. Given: $N = 676, s^2 = 229^2, Z_\alpha = 1.96$

Solution: Initial sample size ignoring fpc:

$$n_0 = \frac{Z^2 s^2}{d^2} = \frac{1.96^2 \cdot 229^2}{1000^2} \approx 402$$

Finite population adjustment:

$$n = \frac{n_0}{1 + \frac{n_0}{N}} = \frac{402}{1 + \frac{402}{676}} \approx 252$$

Hence, the required sample size is $n \approx 252$.

Cross Sectional Studies:
Key concepts & inputs

Confidence level $(1-\alpha)$ e.g. 95% corresponds to $\alpha = 0.05$ and $Z_{\alpha/2} = 1.96$.

Power (for comparisons) $= 1-\beta$, common choices: 0.8 or 0.9.

Effect size the difference you want to detect (e.g. change in prevalence or mean).

Margin of error (precision) E for estimating a parameter (e.g. $\pm 5\%$ for prevalence).

Variance inputs $p(1-p)$ for proportions; σ^2 (or σ) for continuous outcomes.

1) Estimating a **Prevalence** (proportion) — formula

For a desired margin of error E and confidence level (two-sided):

$$n = \frac{Z_{\alpha/2}^2 \, p(1-p)}{E^2}$$

- If p unknown, use $p = 0.5$ to get the maximum required sample size.
- $Z_{\alpha/2}$ is the standard normal critical value (e.g. 1.96 for 95% CI).

example: Prevalence of dental caries

Goal: estimate the prevalence among dentistry students with 95% CI and a margin of error, $E = 0.05$.

Assume expected prevalence $p = 0.45$ (from prior study/pilot).

Step-by-step arithmetic:

$$Z_{0.025} = 1.96$$

$$1.96^2 = 3.8416$$

$$p(1-p) = 0.45 \times 0.55 = 0.2475$$

$$\text{Numerator} = 3.8416 \times 0.2475 = 0.951036$$

$$E^2 = 0.05^2 = 0.0025$$

$$n = 0.951036 \div 0.0025 = 380.4144$$

$$\Rightarrow \quad n = 381 \text{ (round up)}$$

Adjust for non-response / refusal

If you expect a 10% non-response:

$$n_{final} = \frac{n}{1 - \text{nonresponse fraction}} = \frac{381}{0.90} = 423.333... \Rightarrow 424.$$

Note: why we divide by $1 - f$ (we need the planned sample *after* loss).

2) Estimating a **Mean** — formula

When the outcome is continuous (e.g. DMFT index, plaque score):

$$n = \left(\frac{Z_{\alpha/2}\,\sigma}{E} \right)^2$$

- σ: expected population standard deviation (from pilot or literature).
- E: acceptable margin of error (same units as the mean).

Example: Mean DMFT (Decayed, Missing, and Filled Teeth) score

Goal: 95% CI for mean DMFT with margin of error$E = 0.5$ DMFT points.

Assume $\sigma = 2.5$.

Step-by-step:

$$1.96 \times \sigma = 1.96 \times 2.5 = 4.9$$

$$\frac{4.9}{E} = \frac{4.9}{0.5} = 9.8$$

$$n = 9.8^2 = 96.04 \Rightarrow 97 \text{ (round up)}$$

3) Comparing two **Proportions** (two groups)

Used when comparing prevalence between groups (e.g. males vs females):

$$n = \frac{\left(Z_{\alpha/2}\sqrt{2\bar{p}(1-\bar{p})} + Z_{1-\beta}\sqrt{p_1(1-p_1)+p_2(1-p_2)}\right)^2}{(p_1-p_2)^2}$$

- $\bar{p} = (p_1 + p_2)/2$.
- This yields n per group (assumes equal group sizes).

Worked example: Gingivitis prevalence, male vs female

Goal: detect difference from 30% (males) to 20% (females) ($\Delta = 0.10$), $\alpha = 0.05$, power=0.8.

Steps:

$$\bar{p} = 0.25, \quad Z_{0.025} = 1.96, \; Z_{0.8} = 0.842$$

$$\sqrt{2\bar{p}(1-\bar{p})} = \sqrt{0.375} \approx 0.612372$$

$$\sqrt{p_1(1-p_1) + p_2(1-p_2)} = \sqrt{0.37} \approx 0.608276$$

$$A = 1.96 \times 0.612372 \approx 1.200$$

$$B = 0.842 \times 0.608276 \approx 0.513$$

$$(A+B)^2 \approx 1.713^2 \approx 2.935$$

$$n = 2.935/0.01 = 293.5 \Rightarrow 294 \text{ per group}$$

4) Comparing two **Means** (two groups)

Approximate (equal group sizes, known common SD):

$$n_{\text{per group}} = \frac{2\,(Z_{\alpha/2} + Z_{1-\beta})^2\,\sigma^2}{\Delta^2}$$

- Δ: difference in means to detect.
- Convert to Cohen's $d = \Delta/\sigma$ for intuition.

Worked example: Detect mean difference

Goal: detect difference $\Delta = 3$ points in a clinical index, $\sigma = 10$, 95% CI and 80% power.

$$Z_{0.025} = 1.96,\ Z_{0.8} = 0.842$$

$$(1.96 + 0.842)^2 = 2.802^2 \approx 7.8504$$

Multiply by 2: 15.7008

$$\times \sigma^2 = 15.7008 \times 100 = 1570.08$$

$$\div \Delta^2 = 1570.08/9 \approx 174.45 \Rightarrow 175 \text{ per group}$$

Design adjustments (practical considerations)

- **Non-response / dropout:** inflate by dividing by $1 - f$ where f is expected loss fraction.
- **Cluster sampling / design effect:** multiply by design effect $D = 1 + (m - 1)\rho$ where m is cluster size and ρ is ICC.
- **Finite population correction (small pop):**

$$n_{adj} = \frac{n_0}{1 + \frac{n_0 - 1}{N}}$$

- **Multiple comparisons:** adjust α or use a different error control (increases n).

Appendix: Quick reference formulas

- Prevalence (proportion): $n = \dfrac{Z^2_{\alpha/2} p(1-p)}{E^2}$
- Mean: $n = \left(\dfrac{Z_{\alpha/2}\sigma}{E} \right)^2$
- Two proportions: see slide with full formula (per-group)
- Two means (per-group): $n = \dfrac{2\left(Z_{\alpha/2} + Z_{1-\beta}\right)^2 \sigma^2}{\Delta^2}$

Try:

1. Compute sample size to estimate prevalence $= 0.1$ with E=0.03, 95% CI.
2. Compare two proportions: p1=0.4, p2=0.25, $\alpha = 0.05$, power=0.9. Find sample per group.
3. For a continuous outcome with $\sigma = 4$ and desired E=0.4, find required n.

(Solutions available on request.)

1) Prevalence (p=0.1, E=0.03, 95% CI)

Formula:

$$n = \frac{Z^2_{\alpha/2}\, p(1-p)}{E^2}$$

Calculation:

$$\begin{aligned} Z^2_{0.025} &= 1.96^2 = 3.8416 \\ p(1-p) &= 0.1 \times 0.9 = 0.09 \\ \text{Numerator} &= 3.8416 \times 0.09 = 0.345744 \\ E^2 &= 0.03^2 = 0.0009 \\ n &= 0.345744 \div 0.0009 = 384.16 \\ \Rightarrow \quad n &= 385 \end{aligned}$$

2) Two proportions (p1=0.4, p2=0.25, power=0.9)

steps:

solns cont.

$A \approx 1.96 \times 0.66238 = 1.298$
$B \approx 1.2816 \times 0.65383 = 0.838$
$(A+B)^2 \approx 4.563, \quad (p_1 - p_2)^2 = 0.0225$
$n_{\text{per group}} \approx 4.563 \div 0.0225 = 202.8$
$\Rightarrow \quad n_{\text{per group}} = 203$

3) Mean ($\sigma = 4$, $E = 0.4$, 95% CI)

$$1.96 \times \sigma = 1.96 \times 4 = 7.84$$
$$\frac{7.84}{E} = 7.84 \div 0.4 = 19.6$$
$$n = 19.6^2 = 384.16$$

Minimum Sample Size

Table 1. Estimating a population proportion with specified absolute precision

$$n = z^2_{1-\alpha/2} P(1-P)/d^2$$

(*a*) Confidence level **95%**

d \ *P*	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
0.01	1825	3457	4898	6147	7203	8067	8740	9220	9508	9604	9508	9220	8740	8067	7203	6147	4898	3457	1825
0.02	456	864	1225	1537	1801	2017	2185	2305	2377	2401	2377	2305	2185	2017	1801	1537	1225	864	456
0.03	203	384	544	683	800	896	971	1024	1056	1067	1056	1024	971	896	800	683	544	384	203
0.04	114	216	306	384	450	504	546	576	594	600	594	576	546	504	450	384	306	216	114
0.05	73	138	196	246	288	323	350	369	380	384	380	369	350	323	288	246	196	138	73
0.06	51	96	136	171	200	224	243	256	264	267	264	256	243	224	200	171	136	96	51
0.07	37	71	100	125	147	165	178	188	194	196	194	188	178	165	147	125	100	71	37
0.08	29	54	77	96	113	126	137	144	149	150	149	144	137	126	113	96	77	54	29
0.09	23	43	60	76	89	100	108	114	117	119	117	114	108	100	89	76	60	43	23
0.10	18	35	49	61	72	81	87	92	95	96	95	92	87	81	72	61	49	35	18
0.11	15	29	40	51	60	67	72	76	79	79	79	76	72	67	60	51	40	29	15
0.12	13	24	34	43	50	56	61	64	66	67	66	64	61	56	50	43	34	24	13
0.13	11	20	29	36	43	48	52	55	56	57	56	55	52	48	43	36	29	20	11
0.14	9	18	25	31	37	41	45	47	49	49	49	47	45	41	37	31	25	18	9
0.15	8	15	22	27	32	36	39	41	42	43	42	41	39	36	32	27	22	15	8
0.20	5	9	12	15	18	20	22	23	24	24	24	23	22	20	18	15	12	9	5
0.25	*	6	8	10	12	13	14	15	15	15	15	15	14	13	12	10	8	6	*

*Sample size less than 5.

Table 1 (*continued*)

(*b*) Confidence level **90%**

d \ *P*	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
0.01	1285	2435	3450	4330	5074	5683	6156	6494	6697	6765	6697	6494	6156	5683	5074	4330	3450	2435	1285
0.02	321	609	863	1082	1268	1421	1539	1624	1674	1691	1674	1624	1539	1421	1268	1082	863	609	321
0.03	143	271	383	481	564	631	684	722	744	752	744	722	684	631	564	481	383	271	143
0.04	80	152	216	271	317	355	385	406	419	423	419	406	385	355	317	271	216	152	80
0.05	51	97	138	173	203	227	246	260	268	271	268	260	246	227	203	173	138	97	51
0.06	**36**	**68**	**96**	**120**	**141**	**158**	**171**	**180**	**186**	**188**	**186**	**180**	**171**	**158**	**141**	**120**	**96**	**68**	**36**
0.07	**26**	**50**	**70**	**88**	**104**	**116**	**126**	**133**	**137**	**138**	**137**	**133**	**126**	**116**	**104**	**88**	**70**	**50**	**26**
0.08	**20**	**38**	**54**	**68**	**79**	**89**	**96**	**101**	**105**	**106**	**105**	**101**	**96**	**89**	**79**	**68**	**54**	**38**	**20**
0.09	**16**	**30**	**43**	**53**	**63**	**70**	**76**	**80**	**83**	**84**	**83**	**80**	**76**	**70**	**63**	**53**	**43**	**30**	**16**
0.10	**13**	**24**	**35**	**43**	**51**	**57**	**62**	**65**	**67**	**68**	**67**	**65**	**62**	**57**	**51**	**43**	**35**	**24**	**13**
0.11	11	20	29	36	42	47	51	54	55	56	55	54	51	47	42	36	29	20	11
0.12	9	17	24	30	35	39	43	45	47	47	47	45	43	39	35	30	24	17	9
0.13	8	14	20	26	30	34	36	38	40	40	40	38	36	34	30	26	20	14	8
0.14	7	12	18	22	26	29	31	33	34	35	34	33	31	29	26	22	18	12	7
0.15	6	11	15	19	23	25	27	29	30	30	30	29	27	25	23	19	15	11	6
0.20	*	**6**	**9**	**11**	**13**	**14**	**15**	**16**	**17**	**17**	**17**	**16**	**15**	**14**	**13**	**11**	**9**	**6**	*
0.25	*	*	**6**	**7**	**8**	**9**	**10**	**10**	**11**	**11**	**11**	**10**	**10**	**9**	**8**	**7**	**6**	*	*

*Sample size less than 5.

Table 2. Estimating a population proportion with specified relative precision

$$n = z^2_{1-\alpha/2}(1-P)/\varepsilon^2 P$$

(*a*) Confidence level **95%**

ε \ P	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
0.01	729904	345744	217691	153664	115248	89637	71344	57624	46953	38416	31431	25611	20686	16464	12805	9604	6779	4268	2022
0.02	182476	86436	54423	38416	28812	22409	17836	14406	11738	9604	7858	6403	5171	4116	3201	2401	1695	1067	505
0.03	81100	38416	24188	17074	12805	9960	7927	6403	5217	4268	3492	2846	2298	1829	1423	1067	753	474	225
0.04	45619	21609	13606	9604	7203	5602	4459	3602	2935	2401	1964	1601	1293	1029	800	600	424	267	126
0.05	29196	13830	8708	6147	4610	3585	2854	2305	1878	1537	1257	1024	827	659	512	384	271	171	81
0.06	**20275**	**9604**	**6047**	**4268**	**3201**	**2490**	**1982**	**1601**	**1304**	**1067**	**873**	**711**	**575**	**457**	**356**	**267**	**188**	**119**	**56**
0.07	**14896**	**7056**	**4443**	**3136**	**2352**	**1829**	**1456**	**1176**	**958**	**784**	**641**	**523**	**422**	**336**	**261**	**196**	**138**	**87**	**41**
0.08	**11405**	**5402**	**3401**	**2401**	**1801**	**1401**	**1115**	**900**	**734**	**600**	**491**	**400**	**323**	**257**	**200**	**150**	**106**	**67**	**32**
0.09	**9011**	**4268**	**2688**	**1897**	**1423**	**1107**	**881**	**711**	**580**	**474**	**388**	**316**	**255**	**203**	**158**	**119**	**84**	**53**	**25**
0.10	**7299**	**3457**	**2177**	**1537**	**1152**	**896**	**713**	**576**	**470**	**384**	**314**	**256**	**207**	**165**	**128**	**96**	**68**	**43**	**20**
0.15	3244	1537	968	683	512	398	317	256	209	171	140	114	92	73	57	43	30	19	9
0.20	1825	864	544	384	288	224	178	144	117	96	79	64	52	41	32	24	17	11	5
0.25	1168	553	348	246	184	143	114	92	75	61	50	41	33	26	20	15	11	7	*
0.30	811	384	242	171	128	100	79	64	52	43	35	28	23	18	14	11	8	5	*
0.35	596	282	178	125	94	73	58	47	38	31	26	21	17	13	10	8	6	*	*
0.40	**456**	**216**	**136**	**96**	**72**	**56**	**45**	**36**	**29**	**24**	**20**	**16**	**13**	**10**	**8**	**6**	*	*	*
0.50	**292**	**138**	**87**	**61**	**46**	**36**	**29**	**23**	**19**	**15**	**13**	**10**	**8**	**7**	**5**	*	*	*	*

*Sample size less than 5.

Table 2 (*continued*)

(*b*) Confidence level **90%**

ε \ P	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
0.01	514145	243542	153341	108241	81181	63141	50255	40590	33074	27060	22140	18040	14571	11597	9020	6765	4775	3007	1424
0.02	128536	60886	38335	27060	20295	15785	12564	10148	8268	6765	5535	4510	3643	2899	2255	1691	1194	752	356
0.03	57127	27060	17038	12027	9020	7016	5584	4510	3675	3007	2460	2004	1619	1289	1002	752	531	334	158
0.04	32134	15221	9584	6765	5074	3946	3141	2537	2067	1691	1384	1128	911	725	564	423	298	188	89
0.05	20566	9742	6134	4330	3247	2526	2010	1624	1323	1082	886	722	583	464	361	271	191	120	57
0.06	**14282**	**6765**	**4259**	**3007**	**2255**	**1754**	**1396**	**1128**	**919**	**752**	**615**	**501**	**405**	**322**	**251**	**188**	**133**	**84**	**40**
0.07	**10493**	**4970**	**3129**	**2209**	**1657**	**1289**	**1026**	**828**	**675**	**552**	**452**	**368**	**297**	**237**	**184**	**138**	**97**	**61**	**29**
0.08	**8034**	**3805**	**2396**	**1691**	**1268**	**987**	**785**	**634**	**517**	**423**	**346**	**282**	**228**	**181**	**141**	**106**	**75**	**47**	**22**
0.09	**6347**	**3007**	**1893**	**1336**	**1002**	**780**	**620**	**501**	**408**	**334**	**273**	**223**	**180**	**143**	**111**	**84**	**59**	**37**	**18**
0.10	**5141**	**2435**	**1533**	**1082**	**812**	**631**	**503**	**406**	**331**	**271**	**221**	**180**	**146**	**116**	**90**	**68**	**48**	**30**	**14**
0.15	2285	1082	682	481	361	281	223	180	147	120	98	80	65	52	40	30	21	13	6
0.20	1285	609	383	271	203	158	126	101	83	68	55	45	36	29	23	17	12	8	*
0.25	823	390	245	173	130	101	80	65	53	43	35	29	23	19	14	11	8	5	*
0.30	571	271	170	120	90	70	56	45	37	30	25	20	16	13	10	8	5	*	*
0.35	420	199	125	88	66	52	41	33	27	22	18	15	12	9	7	6	*	*	*
0.40	**321**	**152**	**96**	**68**	**51**	**39**	**31**	**25**	**21**	**17**	**14**	**11**	**9**	**7**	**6**	*	*	*	*
0.50	**206**	**97**	**61**	**43**	**32**	**25**	**20**	**16**	**13**	**11**	**9**	**7**	**6**	**5**	*	*	*	*	*

*Sample size less than 5.

Table 3. Hypothesis tests for a population proportion

For a one-sided test

$$n=\{z_{1-\alpha}\sqrt{[P_0(1-P_0)]}+z_{1-\beta}\sqrt{[P_a(1-P_a)]}\}^2/(P_0-P_a)^2.$$

For a two-sided test

$$n=\{z_{1-\alpha/2}\sqrt{[P_0(1-P_0)]}+z_{1-\beta}\sqrt{[P_a(1-P_a)]}\}^2/(P_0-P_a)^2.$$

(*a*) Level of significance **5%**, power **90%**, **one-sided** test

P_a \ P_0	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
0.05		239	76	40	25	18	13	10	8	6	5	*	*	*	*	*	*	*	*
0.10	221		378	109	54	33	22	16	12	10	8	6	5	*	*	*	*	*	*
0.15	67	362		498	137	66	39	26	19	14	11	8	7	5	*	*	*	*	*
0.20	34	102	485		601	161	75	44	29	20	15	11	9	7	5	*	*	*	*
0.25	21	49	131	589		686	180	83	48	31	21	16	12	9	7	5	*	*	*
0.30	**15**	**30**	**62**	**156**	**676**		**754**	**195**	**88**	**50**	**32**	**22**	**16**	**12**	**9**	**7**	**5**	*	*
0.35	**11**	**20**	**36**	**72**	**176**	**746**		**804**	**205**	**92**	**52**	**33**	**22**	**16**	**11**	**8**	**6**	**5**	*
0.40	**8**	**14**	**24**	**42**	**80**	**191**	**799**		**837**	**211**	**93**	**52**	**32**	**22**	**15**	**11**	**8**	**6**	*
0.45	**7**	**11**	**17**	**27**	**46**	**87**	**203**	**834**		**853**	**213**	**93**	**51**	**31**	**21**	**14**	**10**	**7**	*
0.50	**5**	**9**	**13**	**19**	**30**	**49**	**91**	**210**	**852**		**852**	**210**	**91**	**49**	**30**	**19**	**13**	**9**	**5**
0.55	*	7	10	14	21	31	51	93	213	853		834	203	87	46	27	17	11	7
0.60	*	6	8	11	15	22	32	52	93	211	837		799	191	80	42	24	14	8
0.65	*	5	6	8	11	16	22	33	52	92	205	804		746	176	72	36	20	11
0.70	*	*	5	7	9	12	16	22	32	50	88	195	754		676	156	62	30	15
0.75	*	*	*	5	7	9	12	16	21	31	48	83	180	686		589	131	49	21
0.80	*	*	*	*	**5**	**7**	**9**	**11**	**15**	**20**	**29**	**44**	**75**	**161**	**601**		**485**	**102**	**34**
0.85	*	*	*	*	*	**5**	**7**	**8**	**11**	**14**	**19**	**26**	**39**	**66**	**137**	**498**		**362**	**67**
0.90	*	*	*	*	*	*	**5**	**6**	**8**	**10**	**12**	**16**	**22**	**33**	**54**	**109**	**378**		**221**
0.95	*	*	*	*	*	*	*	*	**5**	**6**	**8**	**10**	**13**	**18**	**25**	**40**	**76**	**239**	

*Sample size less than 5.

Figure: Hypothesis tests for a population proportion

References & further reading

- Lwanga SK, Lemeshow S. Sample Size Determination in Health Studies. WHO (1991).
- Hulley SB, Cummings SR, et al. Designing Clinical Research. (Chapters on sample size).
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